

Medical Report Simplification System: Enhancing Healthcare Accessibility using NLP-based Extraction and AI-Driven Explanation

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Abstract— Medical reports are not easy to understand for patients, due to their complicated medical jargon and terminology. This can cause misunderstandings, making it confusing and prolonging it harder for the people to track their health conditions without guidance from the professionals. To address this issue, an AI (Artificial Intelligence) powered system for the simplification of Medical Report which extracts difficult medical terms and gives explanations that are more user-friendly. The system uses Named Entity Recognition (NER) with SciSpaCy and UMLS (Unified Medical Language System) to identify the medical terms. And then an AI-based explanation generation model powered by Together.Ai's API (Application Programming Interface) is used to ensure clarity and readability. This initiative, aims to close the communication gap between healthcare providers and patients by turning difficult medical reports into clear explanation, ultimately enabling users to make well-informed health choices.

Index Terms— AI-driven, jargon, Named Entity Recognition (NER), SciSpaCy, Together.ai, UMLS, user-friendly.

I. INTRODUCTION

Medical reports are essential documents that communicate a patient's health status, diagnostic results, and treatment strategies. However, these reports mostly include complicated medical jargon, abbreviations, and medical language that can be hard for those without a medical background to understand. This lack of clarity leads to misunderstandings, increased patient anxiety, and will lead the people to depend on healthcare professionals for explanation. Making sure that patients can understand their medical reports is necessary for decision-making, better commitment to treatment plans, and overall health literacy. Hence, a solution that simplifies medical reports by maintaining the accuracy and meaning is necessary. The system uses Natural Language Processing (NLP) techniques to find difficult medical terms present in the report. After finding the difficult medical terms, an AI-powered explanation module will generate simplified descriptions.

In addition, our system includes a multilingual translation feature that allows patients to receive explanations in their comfortable language. This improvement makes sure improved accessibility for non-English speakers by allowing

them to fully understand their medical conditions without language difficulty. By transforming difficult medical reports into user-friendly explanations and multilingual summaries, this system improves healthcare accessibility and empowers patients to understand their health conditions more conveniently. The proposed approach has potential applications in Electronic Health Record (EHR) integration, and patient education, therefore finally contributing to a more informed patient community.

A. Related Work

A few studies were implemented on AI-based methods for simplifying medical texts and aimed to improve the readability for non-expert users. Techniques such as Named Entity Recognition (NER), transformer-based language models and ontology-based, approaches to extract difficult medical terms and generates explanations which is readable to users which have been studied by researchers for a significant period of time. The following works broaden the work done on medical text simplification, describing approaches that align with the goals of this study.

Zhou et al. [1] performed a review of Natural Language Processing (NLP) applications in smart healthcare, focusing on how NLP can be combined with electronic health records (EHRs) to improve the data access and the patient care. Li et al. [2] performed a complete review on Neural Natural Language Processing (NLP) for understanding unstructured data in electronic health records (EHRs), highlighting the improvements in deep learning techniques and their applications in medical text processing. Devaraj et al. [3] labelled the challenge of simplifying medical texts by introducing a new collection of parallel texts containing technical and user-friendly summaries of the biomedical literature. They proposed a novel metric using a masked language models that are trained on some scientific texts to differentiate between the technical and simplified summaries more effectively. Flores et al. [4] explored the methods to enhance the readability of medical text simplification. They did this by introducing an uncertain loss that encourages the

simpler term generation and a reranked beam search decoding technique which is improved for simplicity. This approach showed that there was a improvement in readability across various datasets. Hossain et al. [5] conducted a review on the use of Natural Language Processing (NLP) in Electronic Health Records (EHRs) for healthcare decision-making. They analyzed many Machine Learning (ML) and Deep Learning (DL) techniques by focusing on challenges such as the lack of annotated data and automated tools and by finding the opportunities for improved clinical insights. Li et al. [6] explored the Large Language Models (LLMs) and control mechanisms to improve the readability of biomedical abstract. Van et al. [7] proposed the AutoMeTS, an autocomplete-based system for simplification of medical text. They introduced a parallel dataset of aligned English Wikipedia and a Simple English Wikipedia sentence and then evaluated four pretrained neural language models (PNLMs). Their ensemble model combining BERT, RoBERTa, XLNet, and GPT-2 achieved a 64.52% word prediction accuracy, showcasing improvements in medical text simplification. Moramarco et al. [8] proposed an approach to make clinical notes more patient-friendly by integrating language models with ontologies. Their method used medical ontologies coupled with common terms to clarify complicated medical language, thereby improving patient understanding of clinical notes. Klavans and Muresan [9] developed DEFINDER, a system for automatically creating glossaries to simplify medical literature. This system focuses on extracting and defining complex terms, which helps non-expert readers better understand the material. Elhadad et al. [10] introduced a comprehensive framework for summarizing medical literature, prioritizing customization to meet various user requirements. Their system aimed to break down and summarize complex medical texts, making them more accessible to a broad audience.

B. Motivation and Justification

Most of the existing medical report analysis systems are designed for doctors and healthcare professionals, which focuses on processing Electronic Health Records (EHR) [11], clinical decision-making, and AI-assisted findings. While these tools are made for medical experts, they don't really help the patients who just wish to understand their own reports. Many existing systems either depend on simple keyword extraction, which often misses context, or uses general NLP models that struggle with medical jargon. As a result, patients are left confused of difficult terms, making it harder for them to understand their health conditions without regularly consulting a doctor. Moreover, these systems generally give raw data instead of transparent and structured explanations, which makes them even less user-friendly for non-experts. The AI-driven Medical Report Simplification System addresses these issues by combining advanced NLP technique and AI-Driven generation with a patient-first approach. Using Named Entity Recognition (NER) with SciSpaCy and UMLS, it accurately extracts key medical terms while Together AI's API generates easy-to-understand

explanations. Apart from the traditional methods that just list out medical terms, this system structures the information into simple and readable summaries. Additionally, with the multilingual feature, users can receive the simplified explanation in their preferred language. This makes the medical information accessible for a large number of audiences. By breaking down difficult medical reports into transparent, natural language summaries, this system empowers patients to take control of their health, reducing dependency on medical professionals for basic clarifications.

C. Outline of the Work

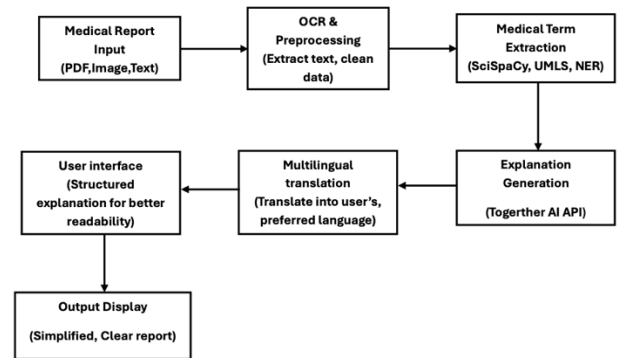


Fig 1. Workflow

Fig. 1 represents the workflow of the Medical Report Simplification System. It shows the interaction between the user and the AI-driven processing components of the system. The input contains medical reports in different formats, including PDFs, images, and raw text, which performs as the primary data sources for the system. The first step in preprocessing where the texts are extracted using a technique called Optical Character Recognition (OCR) from uploaded documents and images. After that, text cleaning and normalization are done to ensure the removal of unnecessary characters, special symbols, and formatting of unusual texts. Once the text is pre-processed, it is passed to the Medical Term Extraction module, Named Entity Recognition (NER) techniques using SciSpaCy and UMLS that identifies key medical entities such as diseases, medications, procedures, and symptoms. This module makes sure that all relevant medical terms are correctly recognized and categorized. After medical term extraction is done, the Explanation Generation module processes the identified terms using Together AI's API model, by generating user-friendly descriptions. This ensures that medical jargon is simplified into understandable language, which enables non-medical background users to understand their reports easily. The last process is Multilingual Translation module, which translates the simplified medical explanations from the previous steps into the user preferred language. This step makes sure that people with limited English proficiency are capable of getting personalized medical insights in their native language. The Output Display presents the fully simplified medical report in an intuitive format, enabling users to review their results.

D. Organization Of The Article

The rest of this article includes Section II which explains the methodologies used in the system for development procedures. Section III tells the details on evaluations, performance metrics and its results. These tests were done to

ensure reliability and accuracy of the system's performance. The impacts of the system are discussed in section IV.

II. METHODOLOGY

The Simplification System for Medical Reports was designed in a systematic and iterative manner to convert complicated medical text into easy-to-understand descriptions for casual users. It starts with OCR and file processing via Tesseract to get text out of multiple formats such as PDFs and images. The text extracted is then preprocessed—cleaning, tokenizing, and removing stop-words—so that it can be made better in quality and appropriate for subsequent analysis. Medical term extraction is subsequently carried out using SciSpaCy and the UMLS knowledge base, allowing for precise identification and categorization of medical entities like diseases, drugs, and procedures. Early efforts to create explanations with transformer-based models such as T5 and domain-adversarial NLP models such as BioBERT were unsuccessful because of problems such as incoherent results, slow execution, and poor adaptability. Consequently, a method based on AI through the Together.AI API was implemented, providing user-friendly, context-aware explanations via optimized prompt engineering. This approach was efficient, precise, and appropriate for practical applications. The system also supports users with various linguistic backgrounds through multilingual translation with the MyMemory API, with the text translated in 500-character blocks for better contextual relevance in Tamil, Hindi, German, and Malayalam. The frontend, which was developed using HTML, CSS, and JavaScript, offers a user-friendly interface through which users can upload medical reports, see concise explanations, and choose preferred languages. In general, the system integrates well AI, medical NLP, and inclusive UI design to provide an extensible, dependable, and user-driven solution for the interpretation of medical reports.

III. EXPERIMENTAL RESULTS AND DISCUSSION

This section explains how the system was evaluated in terms of accuracy, speed, and user satisfaction. It describes the test data, evaluation metrics, and general results showing the medical report simplification system is effective.

A. Dataset Description

A combination of real-world and custom-generated medical reports were used for evaluating the system:

- **Manually Created Reports:** Sample medical reports were written using organised forms and identified medical terms. This made it possible to train and evaluate the explanation test the explanation and extraction models in a safe environment.
- **Online Sample Reports:** Some medical records were collected from open-access sources to show the system to a varied formats and terminologies.

B. Performance Metric Calculation

1) Medical Term Extraction (F1-Score):

The terms extracted by the system against the manually labelled in the dataset were compared to test how well the system could identify medical terms.

Based on that, we calculated:

$$\text{Precision} = \frac{\text{Correct terms extracted}}{\text{Total terms extracted}} \quad (1)$$

$$\text{Recall} = \frac{\text{Correct terms extracted}}{\text{Total actual terms}} \quad (2)$$

$$\text{F1 - Score} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (3)$$

Final F1-score = 91.5%

This score showed that the model could reliably extract relevant medical terms.

2) Explanation Quality:

The explanation generation model was tested by generating simplified outputs for about 50 reports. These were manually reviewed by us and a few medical students to check if:

- The explanation matched the report's content.
- It could be understood without a medical background.
- Around 89% of the explanations were found to be understandable and mostly accurate.

3) Translation Consistency:

To test the translation feature, 20 simplified explanations were selected and they were translated to Tamil, Hindi, and German. Manual checking was done to ensure that the output stay true to the meaning.

- About 85% of the translations preserved the original meaning without confusion.

4) Response Time

The time taken by the model to produce response was also tested. Using basic time tracking, the time taken to generate explanations were recorded in 10 different cases.

- The average response time was between 3–5 seconds, which is acceptable for real-time use.

C. Metric Results Summary

The performance metrics results on various testing done are shown in the below table I.

TABLE I. PERFORMANCE OF METRICS

Metric	Evaluation method	Results
Medical Term Extraction	F1-score from manual comparison	91.5%
Explanation Quality	Manual review & user understanding	89% Accuracy
Translation Consistency	Manual check of meaning preservation	85%
Response Time (AI-generated)	Time tracking across 10 test queries	3-5 Seconds

D. Model Comparison

Many models were taken into considerations as we discussed in section II B, and the best solution is taken as final

proposed model after experimentations. Model's comparison based on accuracy and other factors are shown in table II

TABLE II. MODEL COMPARISON

Aspect	Positive Feedback	Remarks
Ease of Understanding	92%	Users found explanations simple to read
Accuracy of Summaries	89%	Mostly correct according to users
Translation Clarity	85	Slight issues in some languages
Overall User Satisfaction	90%	Users were happy with the system output

E. User Feedback

Nearly 20 users (10 medical, 10 non-medical) were asked to test the system and their feedback was collected using a short form with checkboxes for clarity, usefulness, and translation quality which is summarized in table III.

In the evaluation, it confirms that the system effectively extracts, simplifies, and translates medical reports. It performs fast enough for real-time use. Minor improvements in translation clarity can further enhance system performance.

TABLE III. USER FEEDBACK SUMMARY

Model	Accuracy	Response Time	Remarks
T5 Transformer	<1%	1-3 minutes	Irrelevant or gibberish output
BioBERT + NLP Rules	~20%	~12 seconds	Missed context, rigid output
AI-Based (Together.AI)(Proposed*)	89%	3-5 seconds	human-like simplification

IV. CONCLUSION

In this study, a Medical Report Simplification System was developed that helps the user to upload the medical report for a clear and user-friendly explanation. The details of the processing techniques and methodologies were discussed in section II, where a detailed description of processing steps are given. Also, we discussed about the challenges that were faced during the selection of the right model for explanation generation. Transformer based models and rule-based NLP models were considered and implemented but those models did not give expected and accurate results. They required more time and training. After a long struggle we found a new approach which is AI-Driven and it completely worked. Next, we moved on to the final module translation part. Initially, using google translation API was considered but eventually dropped due to unofficial API as it can stop working at any time. So, alternatives were explored and we got MyMemory translation API. Since we focus on regional languages this API was opted. In case, in future if there is a plan to expand the lingual diversity we may deploy more advanced translation APIs. Moving on further, At last performance metrics were calculated as we discussed in section III by involving model comparison and user feedbacks. Hence deploying the best performing models into our system to ensure user satisfaction, clarity and readability was done. This system will bridge the gap between difficult medical jargons

and non-medical background people thereby enhancing health knowledge literacy. Making the medical reports more accessible to everyone. In future, we may deploy chatbots for recommendation and doubt clearing purpose making the system more efficient.

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